

Towards Few-Shot Training-Free Anomaly Sound Detection

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Abstract

Anomaly sound detection is an important audio task, with various applications in industrial monitoring, healthcare, security and surveillance, and other domains. Existing methods are often designed with various assumptions for specific types of anomalies and are not suitable when applied to the challenging real-world scenarios, such as domain shifts and data scarcity issues. In this work, we propose a few-shot training-free method, leveraging pre-trained audio models to extract patch-based spatial-temporal representations for few-shot anomaly detection and segmentation. We show that our proposed approach is flexible and applicable even under the extreme low-shot regimes, and can at times outperform models trained with the full datasets. Furthermore, our method is more robust in tackling domain shifts, with the need of only few-shot data points to quickly adapt to various conditions, therefore more suitable for deployment in real-world applications.

Index Terms: anomaly sound detection, machine condition monitoring, few-shot learning, training-free methods

1. Introduction

Anomaly sound detection (ASD) is an important and practical task with numerous applications, including monitoring machine operating conditions, conducting end-of-line testing in manufacturing pipelines [1–4], and healthcare [5, 6]. Audio can be a complementary or alternative sensory modality that is cost-effective, non-contact and ubiquitous (unlike, e.g. vibration, torque, temperature or most bio-sensing) and robust to low visibility and occlusion (unlike, e.g., vision).

Conventional reconstruction-based [7, 8] methods usually utilize an *autoencoder* (AE) architecture, requiring certain number of normal samples to fit a distribution to identify anomaly outliers. In the field of vision, recent generative models like GANs and diffusion models are also utilized to reconstruct normal data distribution for anomaly detection [9, 10]. These methods require lots of domain-specific data and fail to generalize to new domains. Deviation-based methods (e.g., DevNet [11]) relax the constraint of only training with normal samples, mixing few anomaly samples with larger normal sets to obtain better results. Furthermore, clustering [12, 13] and classification-based techniques [14, 15] turn the original outlier detection problem into a binary classification task, treating normal samples from different sources (e.g., different machines) as anomalies to establish a more compact decision boundary. This approach typically requires significant amounts of labeled data. There is also growing interest in leveraging *large language models* (LLMs) toward building foundation models for anomaly detection, es-

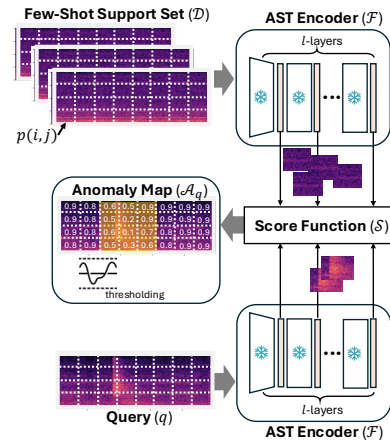


Figure 1: The proposed few-shot patch-based difference framework.

pecially in vision [16–19]. However, it is challenging to apply such methods directly to audio, as common natural language descriptions of sound anomalies are not well defined and broadly available. Also, unlike vision, the background component is not occluded by the source of the anomaly, but additively combined. Thus, few shot solutions, that can easily adapt to new domains with very little data, are especially appealing in audio.

As highlighted in the discussion and results for Task 2 of the 2024 DCASE challenge [20, 21], existing approaches are often impractical and unfeasible for real-world applications due to data scarcity issues. For example, in the case of machine anomaly detection in industrial applications, the performance of models may decrease for machine types not seen in the training data. Also, the assumption of collecting normal and particularly abnormal samples from every single machine type is impractical [22]. In such scenarios, pre-trained audio models on large-scale data are often utilized to address generalization issues across various data distributions [23–25]. Therefore, few-shot ASD approaches have recently become important research topics [26–30]. These works leverage few-shot learning [31, 32] for adapting pre-trained normal models to new domains with few anomaly samples.

In this work, we introduce a patch-based *difference* method for the ASD task. We demonstrate the effectiveness and generalizability of our proposed methods across diverse datasets including machine anomaly and healthcare domains. Our contributions are: 1) We propose a few-shot, training-free ASD approach that requires minimal data samples and is applicable to a wide range of real-world conditions. 2) Our approach demonstrates robustness to domain-shifts and cross-domain general-

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ization in both industrial and medical data, representing a step forward toward a foundational solution for ASD.

2. Few-shot Patch-based Difference Method

We aim to rapidly adapt to domain shifts (e.g., changes in machine configurations) without requiring extensive data collection and model fine-tuning, while maintaining high performance and robustness against variations. Therefore, we formulate the problem as few-shot anomaly detection, relying on only a *limited number of available normal* samples. Our method is inspired by WinCLIP [16], a vision-based anomaly detection framework that computes anomaly scores by leveraging visual differences. Differently, we dispense the language component entirely to account for the non-availability of textual descriptions for audio anomaly, relying solely on intermediate patch-level audio features from a pre-trained transformer encoder. This way, our approach is not tied to a specific encoder, any transformer-based encoder trained on large-scale data can be integrated into our framework. Another closely related work [33] also utilizes hidden representations from a pre-trained audio transformer for anomaly detection. However, their method targets a low-shot setting (approximately 200 or more normal samples), whereas our approach focuses on a more challenging few-shot regime (fewer than 10 normal samples). In this study, we use the *audio spectrogram transformer* (AST) [34] encoder, which is trained based on the *vision transformer* (ViT) architecture [35]. AST is a widely adopted transformer-based general purpose audio encoder in various research studies [36, 37].

Figure 1 illustrates our proposed framework with a training-free inference pipeline focusing on the exposure of local differences in audio spectrogram. Let $\mathcal{D} = \{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_N\}$ be an N -shot normal support set (typically with $N \leq 10$), and $\mathbf{q}$ be a query sample. We first compute a 128-bin log Mel spectrogram for each audio sample, following the configuration from the original AST paper (25 ms window size and 10 ms hop size). Each spectrogram is then divided into P sub-patches, indexed by (i, j) , which we denote as $p(i, j)$. Accordingly, each normal support sample is represented as $\mathbf{x}_n = p_n(i, j)$ while the query sample is represented as $\mathbf{q} = p_q(i, j)$. These sub-patches $p(i, j)$ are passed to a frozen, pre-trained AST encoder (adopted from Huggingface¹). The AST model consists of $L = 11$ transformer layers. At each layer $l \in 1, \dots, L$, its corresponding latent representation extracted by the AST encoder $\mathcal{F}_l(\cdot)$ is divided into 12×101 sub-patches (i.e., $i = 12$ and $j = 101$). Therefore, we can obtain a total of $N \times 12 \times 101$ patch-level representations per layer from the support set. We denote it as the collection set of searching keys $P'_l \in \mathbb{R}^{(12 \times 101 \times N) \times d}$ in Eq. (1), where the AST hidden dimension $d=768$.

$$P'_l = \bigcup_{n=1}^N \mathcal{F}_l(\mathbf{x}_n) \quad (1)$$

To detect anomalies, we compute the cosine distance by calculating the *pairwise cosine similarity* $\mathbf{S}$ between the input query patches $\mathcal{F}_l(\mathbf{q}) \in \mathbb{R}^{12 \times 101 \times d}$ and the searching keys P'_l as shown in Eq. (2), where $\mathbf{S}_{a,b}$ is the cosine similarity between vectors of $\mathcal{F}_l(\mathbf{q})$ and P'_l . We select only the maximum value as the representative score over the P'_l comparisons. We formally define the scoring function $\mathcal{S}(\cdot)$ in Eq. (3), to reveal the most dissimilar regions. Finally, we aggregate these sub-patch-level differences by averaging across all the layers, re-

sulting in an averaged anomaly map $\mathcal{A}_q$ as shown in Eq. (4). Since each transformer layer captures different feature representations (from low-level to high-level), queries with large deviations across multiple layers can be treated as an anomaly. Consequently, if the pooling value of anomaly map $\mathcal{A}_q$ exceeds a certain threshold, the query is considered anomalous. We obtain a single anomaly score by taking a specified quantile of values from this anomaly map, where the quantile acts as a hyperparameter that controls detection sensitivity. We refer to this proposed approach as *AST-PB*.

$$\mathbf{S}_{a,b} = \langle \mathcal{F}_l(\mathbf{q})_a, P'_{lb} \rangle, \quad \mathbf{S} \in \mathbb{R}^{12 \times 12 \times (N \cdot 12 \times 12)} \quad (2)$$

$$\mathcal{S}(\mathcal{F}_l(\mathbf{q}), P'_l) = 1 - \max_b(\mathbf{S}_{:,b}) \quad (3)$$

$$\mathcal{A}_q = \frac{1}{L} \sum_{l=1}^L \mathcal{S}(\mathcal{F}_l(\mathbf{q}), P'_l) \quad (4)$$

3. Experimental Design

Table 1: Data statistics of the used public and private datasets.

Dataset	# Types	# Train	# Test	len (sec)
MIMII	4	400 - 800	200 - 600	10
DCASE-Dev	7	990(SRC), 10(TGT)	100(SRC), 100(TGT)	10
DCASE-Eval	9	990(SRC), 10(TGT)	100(SRC), 100(TGT)	10
Kadia	1	20	36	10
Asthma	1	101	42	1-2

Data: We perform few-shot scenario studies on diverse public and private datasets to showcase the generalizability of the proposed approach across various domains. For this, we consider a total of five datasets (three public and two private) to comprehensively evaluate the model robustness and generalization across different domains and conditions. Table 1 shows the detailed data statistics, namely: **MIMII** [38]: The dataset comprises four machine types recorded at different SNRs and machine IDs; **DCASE2024-Task2-Development (DCASE-Dev)** [38, 39] and **DCASE2024-Task2-Additional (DCASE-Eval)** [21, 39]: These two datasets are from DCASE2024-Task2 Challenges, containing various machine types, recorded in both *source* (SRC) and *target* (TGT) domains, with balanced test set²; **Kadia**: This is a Bosch proprietary dataset collected in the manufacturing line. The anomaly comes from misalignment of the honing tool-tip, causing a squealing sound; **Asthma** [40]: This is another private dataset in the healthcare domain, featuring normal coughing and coughing with wheeze. The samples containing wheezing are classified as anomalies. This data was collected during patients visit to their healthcare provider.

For MIMII and the two private datasets, we follow the partitioning strategy adopted from [38]. To create a balanced test set, we use all of the anomaly samples along with the same number of randomly selected normal samples. The remaining normal data is used for training. Notice that we randomly craft a small subset (i.e., 10 shots of normal and abnormal samples) from MIMII as the validation set to study the role of hyperparameters for the proposed approach (see Sec. 4.1). For the DCASE datasets, we respect the official train and test splits.

Baseline: We consider *autoencoder* (AE) [41] as our baseline method, which is the most conventional approach for benchmarking anomaly detection systems [8, 21]. The AE model is trained to reconstruct normal samples, and the reconstruction

¹https://huggingface.co/docs/transformers/en/model_doc/audio-spectrogram-transformer

²https://github.com/nttclab/dcase2024_task2_evaluator

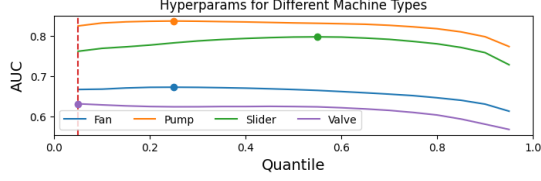


Figure 2: AUC performances of different decision quantiles per machine type based on the crafted small MIMII validation set. The red dashed vertical line represents the default 0.05 quantile.

error is utilized as the anomaly score during inference. It is also commonly assumed that a significant amount of normal data is accessible during the training stage. We directly adopt the official implementations of AE as our baseline system³⁴. Notice that the AE baseline is trained separately for each dataset, utilizing all available in-domain normal data for the training.

Metrics: We follow the same evaluation from DCASE2024-Task2 and report *area under the curve* (AUC) as our main metric, and compute *harmonic mean* (HMean) for the overall aggregation.

Experiments: We conduct experiments with different scenarios by varying the number of normal shots used to train the AE model or for the proposed AST-PB support set. First, *all normal* (denote as all) samples are used from the training set. Second, few-shot $N \in (1, 5, 10)$ (denote as the number) normal samples are randomly selected from the original full training set. All the experiments are conducted in three trials (i.e., with different random seeds), and the mean and *standard deviation* (STD) of the evaluation metrics are reported.

4. Results and Discussions

4.1. Hyperparameter Study

There are two main hyperparameters to consider for the proposed AST-PB method: 1) the choice of the intermediate network layers L (i.e., from 1 to 11) that are used in calculating the anomaly map $\mathcal{A}_q$; and 2) the percentage quantile (e.g., 0.05, 0.1, ..., 0.95) of the anomaly score decision threshold. We use the small crafted MIMII validation set (Sec. 3) to tune these hyperparameters. Firstly, we observe that utilizing all L layers yields the most robust results. Secondly, Figure 2 shows the impact of decision threshold (via percentage quantile) on AUC performance of the validation set. We observe that different machines do require different thresholds to achieve their best performance (see the solid dots per machine). Although we recognize that tuning hyperparameters could improve performance, we adhere to the official benchmark setup in the DCASE Challenge for the rest of the experiments. For fair comparisons, we fix the hyperparameters obtained from MIMII validation knowledge (i.e., all L layers and a 0.05 decision quantile) across all experiments and datasets. We also provide the ideal results as a reference, denoted as AST-PB (10HT), which would be achieved if we were able to further tune the quantile for each machine individually. Here HT stand for *hyperparameter tuning*.

4.2. Cross-Domain Generalizability

In Table 2, we show the AUC results per machine type for MIMII dataset on the left, which is aggregated across different machine IDs and SNRs. We see that the performance of pro-

Table 2: MIMII and the two private datasets results in AUC (averaged over different SNR conditions and machine IDs. $STD \leq 0.05$). **Bold:** best few-shot results. Underline: outperform AE (all).

	MIMII					Private	
	Fan	Pump	Slider	Valve	HMean	Kadia	Asthma
AE (all) [38]	0.790	0.723	0.786	0.613	0.720	0.309	0.217
AE (1)	0.702	0.641	0.611	0.474	0.594	0.428	0.217
AE (5)	0.714	0.700	0.644	0.456	0.608	0.336	0.218
AE (10)	0.716	0.696	0.658	0.454	0.610	0.319	0.220
AST-PB (1)	0.628	0.787	0.726	0.575	0.669	0.421	0.617
AST-PB (5)	0.661	0.828	0.754	0.600	0.700	0.556	0.626
AST-PB (10)	0.671	0.859	0.765	0.613	0.715	0.733	0.635

posed AST-PB (10) is comparable and at times outperforming (e.g. pump) the AE (all) benchmark, which is trained with the entire training samples. Looking at the few-shot performance, our proposed method clearly outperform the AE baselines for the corresponding shots (1, 5, 10). We can also observe that, adding more shots help to improve the performance in general. Table 2 on the right shows the results of two private datasets, Kadia in manufacturing domain and Asthma in healthcare domain. We notice similar trends here as in the MIMII dataset, with a notable performance improvement when using our proposed AST-PB approach compared to AE (all). We attribute this significant performance improvement to the limited availability of normal samples to adequately represent the normal space. Methods such as AE typically require a diverse range of training samples to learn the robust *normal distribution*, whereas AST-PB demonstrates its robustness and effectiveness even with only a few in-domain samples available.

4.3. Cross-Domain Robustness

Table 3 and Table 4 show the detailed results in SRC and TGT domains from DCASE-Dev and DCASE-Eval, respectively. Firstly, looking into the SRC domain results, we observe comparable or sometimes superior performance between the AE (N) results and AST-PB (N). However, for this dataset, AE (all) demonstrates better performance on average. Looking closer, there are several machine types that AST-PB demonstrates better performance, such as GearBox, ToyTrain; while worse at ToyCar. Similar findings hold for both DCASE-Dev and DCASE-Eval sets.

We speculate that the variance in performance may be attributed to the characteristics of the anomaly types (e.g., periodic vs. aperiodic/impulsive, stationary vs. non-stationary), where AST-PB and AE may each have their own advantages in handling different types. It would be an interesting future direction to take different characteristics of anomaly types into consideration and design targeted specific score function (S). Another hypothesis is that different machines or anomaly types require different set of hyperparameters, as shown in Figure 2 as we see a consistent improvement of AST-PB (10HT) results if we are able to perform *hyperparameter tuning* properly.

Additionally, the results from the TGT domain could provide us with a better understanding of the performance of these approaches, as TGT simulates real-world scenarios where machine operation could constantly change, and only a few-shot samples are available (i.e., 10 shots in the DCASE setup) for unsupervised domain adaptation. We note substantial performance improvements of the proposed AST-PB approach over the AE baselines, even when compared to the one trained on the

³https://github.com/nttcs/nttcslab/dcase2023_task2_baseline_ae

⁴https://github.com/MIMII-hitachi/mimii_baseline

Table 3: DCASE-Dev results of source (SRC) and target (TGT) domains in AUC (average over 3 trials, $STD \leq 0.06$).

	Fan		Slider		Valve		ToyTrain		ToyCar		Bearing		GearBox		HMean	
	SRC	TGT	SRC	TGT	SRC	TGT	SRC	TGT	SRC	TGT	SRC	TGT	SRC	TGT	SRC	TGT
AE (all) [21]	0.677	0.552	0.665	0.560	0.511	0.463	0.766	0.469	0.670	0.338	0.620	0.614	0.704	0.693	0.650	0.503
AE (1)	0.577	0.455	0.462	0.582	0.548	0.573	0.385	0.402	0.551	0.538	0.431	0.422	0.290	0.449	0.440	0.479
AE (5)	0.706	0.342	0.54	0.511	0.462	0.408	0.554	0.407	0.508	0.452	0.536	0.564	0.452	0.528	0.527	0.447
AE (10)	0.721	0.295	0.580	0.524	0.442	0.329	0.545	0.429	0.548	0.370	0.539	0.620	0.485	0.550	0.541	0.417
AST-PB (1)	0.571	0.533	0.556	0.540	0.568	0.511	0.469	0.552	0.395	0.528	0.523	0.578	0.577	0.589	0.514	0.546
AST-PB (5)	0.576	0.588	0.555	0.573	0.505	0.500	0.590	0.601	0.420	0.494	0.514	0.613	0.624	0.661	0.533	0.570
AST-PB (10)	0.569	0.576	0.559	0.566	0.502	0.498	0.666	0.626	0.393	0.454	0.516	0.650	0.639	0.691	0.535	0.580
AST-PB (10HT)	0.575	0.636	0.571	0.613	0.665	0.515	0.668	0.630	0.518	0.477	0.548	0.655	0.646	0.712	0.594	0.595
SOTA [24]	0.653	0.656	0.935	0.835	0.790	0.848	0.780	0.627	0.619	0.650	0.703	0.765	0.784	0.755	0.740	0.724

Table 4: DCASE-Eval results of source (SRC) and target (TGT) domains in AUC (average over 3 trials, $STD \leq 0.07$). Abbreviations: 3DPrinter (3DP), AirCompressor (AIR), BrushlessMotor (BRU), HairDryer (HAI), HoveringDrone (HOV), RoboticArm (ROB), Scanner (SCA), ToothBrush (TOO), ToyCircuit (TOY).

	3DP		AIR		BRU		HAI		HOV		ROB		SCA		TOO		TOY		HMean	
	SRC	TGT	SRC	TGT	SRC	TGT	SRC	TGT	SRC	TGT	SRC	TGT	SRC	TGT	SRC	TGT	SRC	TGT	SRC	TGT
AE (all) [21]	0.791	0.480	0.650	0.482	0.707	0.635	0.622	0.460	0.860	0.439	0.660	0.415	0.692	0.537	0.750	0.695	0.766	0.527	0.715	0.506
AE (1)	0.457	0.458	0.468	0.502	0.643	0.462	0.501	0.410	0.184	0.633	0.504	0.506	0.333	0.460	0.786	0.320	0.597	0.261	0.424	0.419
AE (5)	0.536	0.508	0.499	0.455	0.654	0.482	0.679	0.248	0.643	0.272	0.609	0.402	0.348	0.474	0.836	0.288	0.773	0.24	0.583	0.344
AE (10)	0.590	0.502	0.508	0.444	0.749	0.500	0.710	0.252	0.836	0.235	0.582	0.443	0.586	0.522	0.865	0.256	0.788	0.234	0.669	0.336
AST-PB (1)	0.483	0.595	0.537	0.461	0.498	0.581	0.458	0.445	0.673	0.545	0.521	0.593	0.479	0.71	0.599	0.671	0.766	0.654	0.542	0.571
AST-PB (5)	0.509	0.536	0.547	0.464	0.510	0.617	0.393	0.425	0.528	0.594	0.522	0.63	0.671	0.753	0.671	0.720	0.776	0.582	0.550	0.573
AST-PB (10)	0.516	0.498	0.552	0.486	0.533	0.626	0.405	0.408	0.570	0.598	0.530	0.666	0.695	0.773	0.647	0.791	0.765	0.597	0.562	0.580
AST-PB (10HT)	0.560	0.549	0.575	0.528	0.545	0.762	0.537	0.562	0.634	0.688	0.561	0.698	0.751	0.864	0.744	0.886	0.809	0.762	0.621	0.677
SOTA [24]	0.678	0.683	0.657	0.641	0.675	0.712	0.748	0.715	0.695	0.808	0.636	0.854	0.981	0.919	0.744	0.704	0.675	0.678	0.710	0.737

entire dataset containing both source and target domains (see the TGT HMean results for AST-PB (10) and AE (all) in Table 3 and Table 4, values highlighted with bold and underline). This suggests that our approach can rapidly adapt to domain shifts by simply swapping the support set, while still delivering robust performance.

Furthermore, we show the *state-of-the-art* (SOTA) methods [24] from DCASE leaderboard in Table 3 and 4, where they also leverage pre-trained audio encoders. However, their better performance (especially in the TGT domain) comes from fine-tuning with pseudo labels, which requires additional training steps. Using similar pre-training and fine-tuning schemes of audio encoders for the proposed AST-PB approach can be a promising research direction.

4.4. Does Adding More Shots Help?

We found that adding more shots beyond 10 does not seem to benefit AST-PB as shown in Figure 3, the average performance tends to saturate while the STD decreases. We consider that it might be because the comparisons between all the sub-patches between the support sets and query is getting noisier when the number of comparisons grows. This suggests that the quality of support set samples is more favorable than their quantity, with a few representative prototypes being more critical. Another interesting future direction would be to explore intelligent sampling strategies for selecting a better support set, such as importance sampling [42].

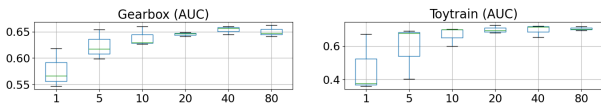


Figure 3: Number of few-shot samples as the support set versus AUC in Gearbox and ToyTrain from DCASE-Dev dataset.

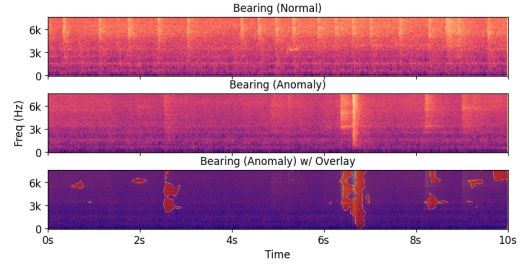


Figure 4: Visualization example of anomaly segmentation.

4.5. Anomaly Sound Segmentation

One additional advantage of the proposed approach is that it can localize spatio-temporal anomalies as we compute anomaly score per sub-patch. The overlay in the bottom of Figure 4 shows the most different patches when compared to the support sets. We can clearly see that the anomaly pattern in about 7s is captured. The overlay particularly emphasizes the energy sweep across the entire frequency region, with these regions contributing significantly to the final anomaly score.

5. Conclusions

In this work, we proposed a few-shot patch-based difference approach for anomaly sound detection and segmentation. Our approach relaxes the assumptions regarding normal and anomaly samples during training, and can adapt to new domains with minimal samples. For future work, we aim to explore sampling methods for selecting support sets and find representative sub-patches for different machine anomalies. We also plan to introduce a data-driven network to predict anomaly scores based on the similarity maps, and design losses for pre-training and fine-tuning AST encoders to further emphasize sample differences.

6. References

- [1] Y. Tagawa, R. Maskeliūnas, and R. Damaševičius, “Acoustic anomaly detection of mechanical failures in noisy real-life factory environments,” *Electronics*, 2021.
- [2] P. Tanuska, L. Spendla, M. Kebisek, R. Duris, and M. Stremy, “Smart anomaly detection and prediction for assembly process maintenance in compliance with industry 4.0,” *Sensors*, 2021.
- [3] H. Chen, Y. Song, Z. Zhuo, Y. Zhou, Y.-H. Li, H. Xue, and I. McLoughlin, “An effective anomalous sound detection method based on representation learning with simulated anomalies,” in *ICASSP*, 2023.
- [4] H. Zhang, J. Guan, Q. Zhu, F. Xiao, and Y. Liu, “Anomalous sound detection using self-attention-based frequency pattern analysis of machine sounds,” in *INTERSPEECH*, 2023.
- [5] T. Fernando, H. Gammulle, S. Denman, S. Sridharan, and C. Fookes, “Deep learning for medical anomaly detection—a survey,” *ACM Computing Surveys (CSUR)*, 2021.
- [6] K. S. Alqudaihi, N. Aslam, I. U. Khan, A. M. Almuhaideb, S. J. Alsunaidi, N. M. A. R. Ibrahim, F. A. Alhaidari, F. S. Shaikh, Y. M. Alsenbel, D. M. Alalharith *et al.*, “Cough sound detection and diagnosis using artificial intelligence techniques: challenges and opportunities,” *IEEE Access*, 2021.
- [7] K. Suefusa, T. Nishida, H. Purohit, R. Tanabe, T. Endo, and Y. Kawaguchi, “Anomalous sound detection based on interpolation deep neural network,” in *ICASSP*, 2020.
- [8] H. Purohit, R. Tanabe, K. Ichige, T. Endo, Y. Nikaido, K. Suefusa, and Y. Kawaguchi, “Mimii dataset: Sound dataset for malfunctioning industrial machine investigation and inspection,” *DCASE Workshop*, 2019.
- [9] A. Jiang, W.-Q. Zhang, Y. Deng, P. Fan, and J. Liu, “Unsupervised anomaly detection and localization of machine audio: A gan-based approach,” in *ICASSP*, 2023.
- [10] Z. Fang, X. Wang, H. Li, J. Liu, Q. Hu, and J. Xiao, “Fastrecon: Few-shot industrial anomaly detection via fast feature reconstruction,” in *CVPR*, 2023.
- [11] G. Pang, C. Shen, and A. Van Den Hengel, “Deep anomaly detection with deviation networks,” in *ACM SIGKDD*, 2019.
- [12] H. Purohit, R. Tanabe, T. Endo, K. Suefusa, Y. Nikaido, and Y. Kawaguchi, “Deep autoencoding gmm-based unsupervised anomaly detection in acoustic signals and its hyper-parameter optimization,” *DCASE Workshop*, 2020.
- [13] R. Müller, F. Ritz, S. Illium, and C. Linnhoff-Popien, “Acoustic anomaly detection for machine sounds based on image transfer learning,” *ICAART*, 2021.
- [14] Y. Koizumi *et al.*, “Description and discussion on dcase2020 challenge task2: Unsupervised anomalous sound detection for machine condition monitoring,” *DCASE Workshop*, 2020.
- [15] P. Primus, “Reframing unsupervised machine condition monitoring as a supervised classification task with outlier-exposed classifiers,” *Tech. Rep., DCASE2020 Challenge*, 2020.
- [16] J. Jeong, Y. Zou, T. Kim, D. Zhang, A. Ravichandran, and O. Dabeer, “Winclip: Zero/few-shot anomaly classification and segmentation,” in *CVPR*, 2023.
- [17] Q. Zhou, G. Pang, Y. Tian, S. He, and J. Chen, “Anomalyclip: Object-agnostic prompt learning for zero-shot anomaly detection,” in *ICLR*, 2024.
- [18] Z. Gu, B. Zhu, G. Zhu, Y. Chen, M. Tang, and J. Wang, “Anomalygpt: Detecting industrial anomalies using large vision-language models,” in *AAAI*, 2024.
- [19] J. Zhu and G. Pang, “Toward generalist anomaly detection via in-context residual learning with few-shot sample prompts,” in *CVPR*, 2024.
- [20] K. Dohi *et al.*, “Description and discussion on dcase 2023 challenge task 2: First-shot unsupervised anomalous sound detection for machine condition monitoring,” *DCASE Workshop*, 2023.
- [21] T. Nishida *et al.*, “Description and discussion on DCASE2024 challenge task 2: First-shot unsupervised anomalous sound detection for machine condition monitoring,” *DCASE Workshop*, 2024.
- [22] N. Harada, D. Niizumi, Y. Ohishi, D. Takeuchi, and M. Yasuda, “First-shot anomaly sound detection for machine condition monitoring: A domain generalization baseline,” in *EUSIPCO*, 2023.
- [23] A. I. Mezza, G. Zanetti, M. Cobos, and F. Antonacci, “Zero-shot anomalous sound detection in domestic environments using large-scale pretrained audio pattern recognition models,” in *ICASSP*, 2023.
- [24] Z. Lv *et al.*, “Aithu system for first-shot unsupervised anomalous sound detection,” *Tech. Rep., DCASE2024 Challenge*, 2024.
- [25] A. Jiang, B. Han, Z. Lv, Y. Deng, W.-Q. Zhang, X. Chen, Y. Qian, J. Liu, and P. Fan, “Anopatch: Towards better consistency in machine anomalous sound detection,” in *INTERSPEECH*, 2024.
- [26] Y. Koizumi, S. Murata, N. Harada, S. Saito, and H. Uematsu, “Sniper: Few-shot learning for anomaly detection to minimize false-negative rate with ensured true-positive rate,” in *ICASSP*, 2019.
- [27] Y. Koizumi, M. Yasuda, S. Murata, S. Saito, H. Uematsu, and N. Harada, “Spidernet: Attention network for one-shot anomaly detection in sounds,” in *ICASSP*, 2020.
- [28] B. Chen, L. Bondi, and S. Das, “Learning to adapt to domain shifts with few-shot samples in anomalous sound detection,” in *ICPR*, 2022.
- [29] Z. Wang, Y. Zhou, R. Wang, T.-Y. Lin, A. Shah, and S. N. Lim, “Few-shot fast-adaptive anomaly detection,” *NeurIPS*, 2022.
- [30] C. Huang, H. Guan, A. Jiang, Y. Zhang, M. Spratling, X. Wang, and Y. Wang, “Few-shot anomaly detection via category-agnostic registration learning,” *IEEE TNNLS*, 2024.
- [31] O. Vinyals, C. Blundell, T. Lillicrap, D. Wierstra *et al.*, “Matching networks for one shot learning,” *NeurIPS*, 2016.
- [32] J. Snell, K. Swersky, and R. Zemel, “Prototypical networks for few-shot learning,” *NeurIPS*, 2017.
- [33] P. Saengthong and T. Shinozaki, “Deep generic representations for domain-generalized anomalous sound detection,” in *ICASSP*, 2025.
- [34] Y. Gong, Y.-A. Chung, and J. Glass, “Ast: Audio spectrogram transformer,” in *INTERSPEECH*, 2021.
- [35] A. Dosovitskiy, “An image is worth 16x16 words: Transformers for image recognition at scale,” in *ICLR*, 2021.
- [36] S. Ghosh *et al.*, “GAMA: A large audio-language model with advanced audio understanding and complex reasoning abilities,” in *EMNLP*, 2024.
- [37] Y. Gong *et al.*, “Listen, think, and understand,” in *ICLR*, 2024.
- [38] K. Dohi, T. Nishida, H. Purohit, R. Tanabe, T. Endo, M. Yamamoto, Y. Nikaido, and Y. Kawaguchi, “MIMII DG: Sound dataset for malfunctioning industrial machine investigation and inspection for domain generalization task,” *DCASE Workshop*, 2022.
- [39] N. Harada, D. Niizumi, D. Takeuchi, Y. Ohishi, M. Yasuda, and S. Saito, “ToyADMOS2: Another dataset of miniature-machine operating sounds for anomalous sound detection under domain shift conditions,” *DCASE Workshop*, 2021.
- [40] S. Ghaffarzadegan, L. Bondi, H.-H. Wu, S. Munir, K. J. Shields, S. Das, and J. Aracri, “Active learning for abnormal lung sound data curation and detection in asthma,” in *INTERSPEECH*, 2023.
- [41] N. Harada, D. Niizumi, D. Takeuchi, Y. Ohishi, and M. Yasuda, “First-shot anomaly detection for machine condition monitoring: A domain generalization baseline,” in *EUSIPCO*, 2023.
- [42] S. T. Tokdar and R. E. Kass, “Importance sampling: a review,” *Wiley Interdisciplinary Reviews: Computational Statistics*, 2010.